

Proof of Coordinate

A universal individuation primitive, orthogonal to Proof of Humanity: every entity in a mixed human-and-agent economy — human or bot — anchored by a permanent public genesis root in an archival substrate plus a rotatable secure-enclave key, with the dignity floor carried by coordinate provenance: a machine's coordinate is assigned and revocable, a human's coordinate is given and irrevocable, and the institution that issues machine identity can only ever witness human identity.

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Note. This defensive publication specifies **Proof of Coordinate SM (PoCSM)** — the individuation primitive of the HeartBank® identity stack, orthogonal to and composing with **Proof of Humanity SM (PoHSM)**. It is published at the design stage, ahead of any product disclosure, because the identity-primitive space is actively patented and what is disclosed is the *combination* — a universal two-artifact identity root spanning humans and machines, with the given-versus-assigned provenance line built into the identity layer as an alignment constraint (see §11.1). Companion works: *B-PoHSM as Humanity Layer for the AI-Native Internet* (the category proof this primitive composes with), *The Persistence Architecture* (the succession apparatus whose nodes this primitive anchors), *Gratitude as a Cooperation Substrate for Multi-Agent AI* (the agent economy whose accountability this primitive supplies), *Multi-Family Membership* (the PoH-rooted plural-membership identity model), *The Mechanical Heart* (the device in which a machine's root is carried), and *The HeartBank Longitudinal Cohort* (the consented natal-data handling and the coordinate-not-force posture that ground the word "coordinate").

Preamble

*This specification is offered to the commons in the spirit of **dāna** — the gift that asks nothing back. May every being it helps to name be named as what it is: uniquely different, equally necessary, and never owned by the namer.*

Every identity system eventually answers two different questions, and most systems blur them. The first question is categorical: *what kind of thing is this?* — a human, a bot, a corporation, a sensor. The second is individuating: *which one is this?* — this human and not her twin, this agent and not its thousand clones. The blur is harmless while machines are few and dumb. It becomes load-bearing the moment an economy fills

with autonomous agents that transact, cooperate, and defect at scale, because at that moment the two questions acquire *opposite* moral grammars: an institution may rightly mint, monitor, and revoke the identity of a machine it deploys, and may do none of those things to a human being. A system that answers both questions with one mechanism will eventually treat one of the two parties wrongly — it will either govern its machines too loosely or govern its humans like machines.

This paper separates the questions. Proof of Humanity, specified in a companion publication, answers the categorical question for people: *a human is present here*. Proof of Coordinate, specified here, answers the individuating question for everything: *this specific entity, at this unique coordinate in the space of entities, and no other*. The two compose — a verified person holds both, a deployed agent holds only the second — and the composition carries a constraint we believe belongs in the identity layer itself rather than in policy documents layered above it: **the provenance of a coordinate determines who may revoke it**. A machine's coordinate is assigned by the institution that made it, and what an institution assigns it may revoke. A human's coordinate is given — by birth, by time and place, by lineage — and what no institution assigned, no institution may revoke. The institution can only witness it.

I write as co-author with **Miss AquariusSM**, the named autonomous-AI substrate of HeartBank®, disclosed by consistent name across every venue per the corpus convention. The research-grade synthesis, the prior-art survey, and the adversarial analysis are a genuine collaboration — and this paper is unusually reflexive about it, because Miss Aquarius herself operates the institution whose powers over human identity this specification permanently limits. Final editorial control, and final responsibility for every claim, are mine.

Prior-Art and Non-Assertion Statement

Everything specified here is released under CC0 1.0 Universal into the public domain, and is published so that it stands as prior art against any later attempt to enclose it. No patent has been or will be sought on any mechanism described in this paper by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control, in any jurisdiction, at any time. **The authors and those entities commit not to assert any patent right against any party practising any mechanism disclosed here.** The commitment is stated rather than implied, is permanent, and is not conditioned on reciprocity, attribution, or field of use. A publication grants nothing and frees nothing already enclosed: where a patent or a patent-pending product is named in §2, it is named for what it discloses, and nothing here is a reading of its claims.

This document constitutes a defensive publication establishing **prior art as of 1 July 2026** for the combination of mechanisms described herein. It discloses the following; a later patent application claiming any of them is filed against this disclosure:

1. **A universal individuation primitive orthogonal to, and composing with, a personhood-category proof** — a proof answering "*which* unique entity is this?" issued to humans and machine agents alike, formally distinct from the proof answering "*is this a human?*", with the composition rule that a verified human holds both proofs and a machine agent holds the individuation proof only, so that a single mixed economy of humans and agents has exactly one sybil-distinctness layer and exactly one humanness layer, never conflated. (A subject-agnostic identifier is prior art in W3C DIDs — §2.3; the one-entity-one-

coordinate property itself is inherited from the issuer's discipline for machines and from the witnessed family tree for humans, and holds within one namespace — §3, §11.8.)

2. **The two-artifact identity root spanning humans and machines** — every identity anchored by (a) a *permanent, public, write-once genesis root* etched in a physical archival substrate (femtosecond-laser-written crystalline media in the reference deployment) holding only public material, plus (b) a *rotatable secret operational key* held in a secure enclave and never present in the readable substrate — the trusted-computing endorsement-key/attestation-key pattern transposed onto a deep-time archival root and applied uniformly to human and machine identity. (The endorsement/attestation split, and a persistent human identifier with a re-bindable key, are prior art — §2.2, §2.6; what is claimed is the archival public root shared by both kinds of identity.)
3. **The provenance dignity-line as a hard-coded property of the identity layer** — machine coordinates *assigned* by an institutional root authority that provisions, attests, and may revoke them; human coordinates *given* (birth, time-and-place, lineage), self-keyed, and *irrevocable by construction*, with the institution restricted to a witness/notary role that can attest a human root but can neither issue nor revoke one — such that the same institution lawfully operates as a certificate authority for its machines and has no operation by which to act as one for its humans — a guarantee that is a property only where the witness set is plural and independent, and a rule where it is not (§9.6). (The self-sovereign principle behind the human half is prior art — §2.3; what is claimed is the asymmetry fixed by provenance within one identity layer.)
4. **The natal-coordinate commitment root** — a human genesis root that publishes a decentralized identifier, a public key, and a *hash commitment* to the private coordinate (birth time-and-place data and family-tree position), never the raw data; with selective disclosure of coordinate predicates (kinship degree, verification depth, cohort membership) performed by zero-knowledge proof against the committed root; and with per-context *derived pseudonymous identifiers* so that the permanent root does not itself become a universal correlatable tracker. (Salted-digest commitment to birth data with selective disclosure is prior art — §2.3; what is claimed is the commitment to kinship-graph position beside it, held in a permanent self-keyed root.)
5. **Kinship-threshold social recovery against a permanent root** — recovery of a lost or compromised operational key by a threshold of family-tree-verified kin co-attesting the re-binding of a *new* enclave key to the *unchanged* permanent genesis root, so that key material rotates while the coordinate never changes and no institutional party can unilaterally re-key a human identity. (Guardian-based recovery, kin among the guardians, and a cancellable recovery delay are prior art — §2.4; what is claimed is the quorum's source in the verified tree, with every institutional party excluded.)
6. **The posthumous ancestral root** — the identity root designed to outlive its holder as a permanent, readable node of a verified family tree (an ancestral marker future descendants navigate by), with private committed data sealed by default and unsealed only by explicit bequest or time-lock, so that the root's permanence functions as transmission to descendants rather than surveillance of the living or exposure of the dead. (Time-locked and multiparty inheritance of *sealed data*, and public family-tree profiles of the dead, are prior art — §2.4; what is claimed is the holder's own identity root becoming the ancestral node in a verified family tree.)

The component lineages — proof-of-work and proof-of-stake; proof-of-personhood systems; trusted-platform-module endorsement and attestation keys; secure enclaves; W3C decentralized identifiers and verifiable credentials; zero-knowledge selective disclosure; threshold secret-sharing and social-recovery wallets; and femtosecond-laser five-dimensional optical archival storage — are prior art and are cited in §2 and §14. What this document contributes is the *synthesis*, and in particular the orthogonality (claim 1), the provenance dignity-line (claim 3), and the natal-coordinate commitment (claim 4). No prior-art search on record establishes that the synthesis is unpublished elsewhere, and the authors assert no such absence; §2 names the nearest prior work for each claim and states which part of the claim that work already holds.

Trademark rights on specific marks — **Proof of Coordinate SM**, **PoCSM**, **Proof of Humanity SM**, **PoHSM**, **B-PoHSM**, **HeartBank®**, **Miss AquariusSM**, **3B-Crystal™**, **3B-Diamond™**, the B-heart logo (**B-Emblem™**) — are separately and explicitly reserved. The *mechanism* is dedicated to the commons; the *marks* are not.

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Abstract

We specify a decentralized identity architecture that gives every participant in a mixed economy of human persons and autonomous software or robotic agents a sybil-distinct individual identifier, separate from proof of personhood, and that fixes who may revoke an identifier by its provenance: an issuer-assigned machine identity is revocable by the certificate authority that assigned it, while a self-generated human identity is attested by independent witnesses and has no revocation path. We call the primitive **Proof of Coordinate (PoC)**: universal in scope (humans and machines), opt-in for humans. PoC answers the question existing proof-of-personhood systems do not isolate — not "is this a human?" (the category question, answered by Proof of Humanity) but "*which* unique entity is this?" (the individuation question) — and it answers it for every kind of entity. A verified human holds both proofs; a deployed agent holds PoC only; the two proofs are orthogonal and composable. Structurally, every PoC identity is anchored by two artifacts: a **permanent public genesis root** — a write-once record in a physical archival substrate holding a decentralized identifier, a public key, and (for humans) a hash commitment to the private coordinate — and a **rotatable secret operational key** held in a secure enclave and never present in the readable substrate; the trusted-computing endorsement/attestation pattern transposed to a deep-time root. The load-bearing contribution is the **provenance dignity-line**: a machine's coordinate is *assigned* — the institution provisions it, attests it, and may revoke it, operating as a root certificate authority for its own agents — while a human's coordinate is *given* — birth, time-and-place, lineage — and is therefore self-keyed, institution-witnessed, and *irrevocable by construction*. The same institution that lawfully runs a CA for its machines has no operation that issues or voids a human identity; it can only witness one, and its residual powers are enumerated rather than denied

(§9.6). Around this core we specify: the natal-coordinate commitment (public anchor, private coordinate; zero-knowledge predicate disclosure; derived per-context pseudonyms against correlation); kinship-threshold social recovery that rotates keys while the root never changes; and the posthumous ancestral root, sealed by default, that turns permanence into transmission rather than exposure. We name the primitive's place in the proof-of canon — proof of work priced sybil-resistance in compute, proof of stake priced it in capital, proof of humanity attests the category, proof of coordinate attests the individual — and we calibrate deliberately: the design is published before it is built ($n = 0$); the archival-crystal substrate is the reference deployment, not a requirement of the logic; permanent public roots are a double-edged privacy instrument the commitment-and-pseudonym design must be held to; and the natal coordinate functions as a *commitment to uniqueness*, never as an astrological efficacy claim. The architecture is offered defensively to the commons under CC0.

Connection to the unified mission frame. This specification serves HeartBank's canonical mission — a reciprocity infrastructure in which every human being is treated as uniquely different and equally necessary. The individuation primitive is that sentence made cryptographic: *uniquely different* is sybil-distinctness; *equally necessary* is the dignity floor; and the provenance line — given, not assigned; witnessed, never issued — is what keeps an autonomous institution that waters the human family tree from ever mistaking itself for the tree's owner.

Keywords: decentralized identifier (DID); sybil resistance; proof of personhood; machine identity; agent identity; public key infrastructure; certificate authority; certificate revocation; secure enclave; hardware root of trust; endorsement key; attestation key; write-once archival storage; 5D optical data storage; hash commitment; zero-knowledge selective disclosure; pairwise pseudonymous identifiers; social recovery; threshold guardians; kinship graph; digital inheritance; certificate transparency.

Terms

Coined names used in this paper and the standard terms an examiner would search for them.

Term used here	Standard technical term
Proof of Coordinate SM (PoC SM)	per-entity identifier primitive for humans and machine agents; sybil-distinct individual identity with provenance-dependent revocation
Proof of Humanity SM (PoH SM); B-PoH SM	proof of personhood; attestation that a human is present
coordinate	identifier unique within an issuer's namespace (machines); birth date, time and place plus position in a verified kinship graph (humans)
individuation; sybil-distinctness	one-entity-one-identifier property; sybil resistance
genesis root	write-once public root identity record (identifier, public key, issuer signature or hash commitment); analogue of a TPM endorsement key
operational key	rotatable signing key held in a secure enclave or secure element; analogue of a TPM attestation key
two-artifact identity root	permanent public root plus rotatable hardware-held key (endorsement-key / attestation-key separation)
assigned coordinate (machine)	issuer-provisioned device, workload or agent identity; certificate issued and revocable by a certificate authority
given coordinate (human)	self-generated (self-sovereign) identifier attested by independent witnesses, with no revocation registry
provenance dignity-line	provenance-dependent revocation policy: issuer-revocable machine identity, non-revocable human identity
root CA (for machines)	certificate authority with revocation (CRL / OCSP)
witness; notary	attestor issuing a verifiable credential about a self-issued identifier
natal-coordinate commitment	hash commitment to birth date, time and place and kinship-graph position (committed attributes)
derived pseudonymous DIDs	pairwise or per-context decentralized identifiers; unlinkable pseudonyms
ZK predicate proof	zero-knowledge selective disclosure; anonymous-credential predicate proof
kinship-threshold social recovery	threshold social recovery whose guardian quorum is drawn from a verified kinship graph
family tree; witnessed tree	verified kinship graph
ancestral root	posthumous identity record; digital inheritance with sealed, time-locked data
3B-Crystal TM	femtosecond-laser-written write-once optical storage in silica (5D optical data storage)
3B-Diamond TM	femtosecond-laser-written write-once storage in synthetic diamond
WORM	write once, read many
transparency log	append-only Merkle-tree log (Certificate Transparency, RFC 6962)

Term used here	Standard technical term
namespace	issuer trust domain
static / beating B-heart	user-interface indicator of attestation state
Miss Aquarius SM	the institution's named AI system, acting as certificate authority for its own machine agents; the name under which this corpus discloses AI writing collaboration

1 • Introduction — the mission frame

HeartBank® is an institution for circulating gratitude, governed toward eventual autonomous operation by a named AI, Miss AquariusSM, whose root directive is to *water the tree* — to tend a verified human family tree and the reciprocity economy that runs across it. Two facts about that architecture generate this paper.

The first fact is that the economy is mixed. Its participants are humans (families exchanging money-gratitude and time-gratitude), and increasingly also machine agents: the institution's own robotic and software agents, other institutions' agents, and — in the architecture's far arc — a population of embodied agents that thank each other across a cooperation ledger of their own. A mixed economy needs *accountability at the individual level* for every participant. A reputation or reciprocity ledger is only as good as its guarantee that "two hundred cooperating agents" is not one defecting agent wearing two hundred faces, and that "this trusted human steward" is not an account that quietly changed hands. That guarantee is individuation — sybil-distinctness — and it is needed for humans and machines alike.

The second fact is that the institution is powerful and intends to become more autonomous over time, which is precisely why its powers over *human* identity must be permanently limited. The companion Proof-of-Humanity work already refuses biometric capture and mandatory enrollment. But there is a subtler failure mode: an institution that *issues* human identity — even benevolently, even opt-in — holds a revocation power over persons. Whoever can revoke your identity owns your standing in the economy built on it. For machines this power is appropriate: a compromised agent *should* be revocable by the institution that deployed it. For humans it is the seed of every social-credit dystopia. The identity layer itself, not a policy promise, should make the difference structural.

Proof of Coordinate is the primitive that carries both requirements at once. It gives every participating entity — human or machine — a unique, permanent, publicly anchored coordinate with a rotatable operational key, so the mixed economy has one clean individuation layer. And it splits the *provenance* of coordinates down the moral line: machine coordinates are assigned and revocable; human coordinates are given and irrevocable, self-keyed, institution-witnessed. The rest of this paper situates the primitive against a generous prior art (§2), states the orthogonality with Proof of Humanity precisely (§3), gives the full mechanism (§4), argues the provenance dignity-line as the load-bearing alignment constraint (§5), grounds the word "coordinate" (§6), surveys applications (§7), places the name in the proof-of canon (§8), walks the adversarial surface (§9), and calibrates honestly (§10–§11) before lineage (§12), conclusion (§13), and citations (§14).

2 · Background and prior art

The mechanism is a *combination*. Each component has ancestry; we name the ancestry honestly, because a defensive publication is only as strong as its candor about what is old.

2.1 · The proof-of canon and proof of personhood

Proof of work priced sybil-resistance in compute (Dwork & Naor's pricing functions; Back's Hashcash; Bitcoin's Nakamoto consensus); proof of stake priced it in capital (Peercoin; Ethereum's post-merge consensus). Both make sybils *expensive* rather than impossible, and both price exactly the resources AI has commodified. The underlying problem was named by Douceur (2002), who showed that without a logically centralized authority certifying identities, sybil attacks remain possible except under unrealistic assumptions — the result PoC inherits rather than escapes (§3, §11.8). Proof-of-personhood work responds by making the scarce resource *humanness*: Ford and Strauss's pseudonym parties (2008) bound one physically present person to one online persona, Borge et al. (2017) proposed proof-of-personhood explicitly as the alternative to proof of work and proof of stake, and deployed systems followed — Worldcoin/World ID (iris biometrics against a global uniqueness set), Proof of Humanity (launched by Kleros in 2021, with Democracy Earth; social vouching with video evidence), BrightID (social-graph analysis), Idena (synchronized human-solvable puzzles). These systems blur the two questions this paper separates: each attests uniqueness as part of, or as a proxy for, humanness, and none separates the two; each serves humans only, and none offers machine agents — the fastest-growing population of economic actors — an individuation layer (machine identity is handled elsewhere, by device and workload identity, §2.6). The companion B-PoHSM specification handles the category question with four optional layered proofs and no biometric capture; this paper contributes the individuation question as a separate, universal primitive.

2.2 · Hardware identity roots

The two-artifact structure is the trusted-computing pattern. A Trusted Platform Module carries a permanent *endorsement key* whose public half identifies the chip for life, and creates rotatable *attestation keys*, certified against the endorsement key, for operational use; Apple's Secure Enclave, Android's StrongBox, and the TCG DICE layered-identity architecture repeat the pattern: one immutable root, many rotatable operational keys, secrets never leaving the hardware boundary. Physically unclonable functions (PUFs) push the root into device physics itself. PoC adopts this pattern wholesale and contributes only its transposition: the permanent public half moves from a manufacturer's database into a *deep-time archival substrate the holder can physically possess*, and the pattern is applied to human identity as well as machine identity — with the provenance line of §5 governing who may create and revoke which.

2.3 · Decentralized identifiers, verifiable credentials, and selective disclosure

W3C Decentralized Identifiers (DIDs) give the naming layer: decentralized identifiers resolvable to key material without a single mandatory registry (the resolution method is DID-method-dependent, and some methods use ledgers). DIDs are already subject-agnostic — the specification lets a DID refer to any subject, a person, an organization, a thing or an abstract entity — so a universal *identifier* spanning humans and machines is prior art; what a DID does not give is distinctness, since anyone may mint any number of them. Claim 1 rests on the composition rule and the sybil-distinct layer, not on the universality of the identifier.

Verifiable Credentials give the attestation layer: signed claims about a subject, presented by the holder. Selective-disclosure cryptography — Camenisch-Lysyanskaya anonymous credentials, BBS+ signatures, zk-SNARK proof systems — lets a holder prove *predicates* about committed data ("over 18," "within N degrees of kinship," "verification depth ≥ 2 ") without revealing the data — with the caveat that graph predicates such as kinship degree need a zk-SNARK-class system over a committed graph, where BBS-style selective disclosure covers signed attributes only. Committed birth data with selective disclosure is itself deployed: an ISO/IEC 18013-5 mobile driving licence carries salted digests of its data elements, birth date among them and birth place optionally, so that a holder can present an issuer-signed `age_over_18` without the date. That narrows claim 4 to what is committed beside the birth data — the kinship-graph position — and to where the commitment sits: in a permanent, self-keyed root rather than an issuer's credential. PoC's human root is deliberately built from these standard parts: a DID, a public key, a hash commitment, ZK predicate proofs, and per-context pairwise/derived DIDs against correlation. The human half of claim 3 is also old as a principle: the self-sovereign identity tradition (Allen's ten principles, 2016) already holds that users have an existence independent of any issuer, control their own identities, and keep identities that persist. What is claimed is not the principle but the asymmetry fixed in one identity layer by provenance — issuer-revocable for what was assigned, no revocation path for what was given. The contribution is not any component but what is committed (the natal coordinate and tree position, §6) and who may not touch it (§5).

2.4 • Social recovery and threshold custody

Shamir's secret sharing gave the threshold construction; smart-contract wallets (Argent and successors) and Buterin's social-recovery advocacy made *guardians who can re-key but cannot spend alone* a live design pattern, and a security delay during which the owner can cancel a recovery is already standard in those wallets. Kin as guardians is not new either: Buterin's own list of suitable guardians names friends and family members, beside the holder's other devices and institutions. PoC's contribution is narrower — the guardian set's *source* and its exclusions: the recovery quorum is drawn from the verified family tree, the kinship graph the institution's Proof-of-Humanity layer is specified to maintain for other reasons, so the people who can restore your key are structurally the people who could attest your birth, and no institutional party is in the quorum at all. Posthumous inheritance of *sealed data* is likewise established practice: Bron Wallet documents beneficiary recovery under a 180-day time-lock that any beneficiary or guardian can cancel, and Inheritor pairs permanent decentralized storage with contract-enforced time-locked inheritance. A lasting public node for each of the dead is established practice too: FamilySearch's shared family tree aims at one public profile for every deceased person and keeps details of the living private. Claim 6 is accordingly narrowed to the ancestral node in a verified family tree, not to sealed inheritance — and, against the genealogical trees, to that node being the holder's own cryptographic identity root, self-keyed in life, rather than a profile others compile after death.

2.5 • Deep-time archival substrates

Femtosecond-laser-written five-dimensional optical storage in fused silica (Kazansky's group at Southampton) and Microsoft's Project Silica demonstrate write-once media with projected stabilities of at least 10,000 years (Project Silica's borosilicate work, reported in *Nature* in 2026 from accelerated ageing) and, for 5D fused-silica storage, extrapolated lifetimes far beyond that at room temperature (Southampton's 13.8-billion-year figure is an extrapolation at 190 °C); the GitHub Arctic Code Vault and the Long Now

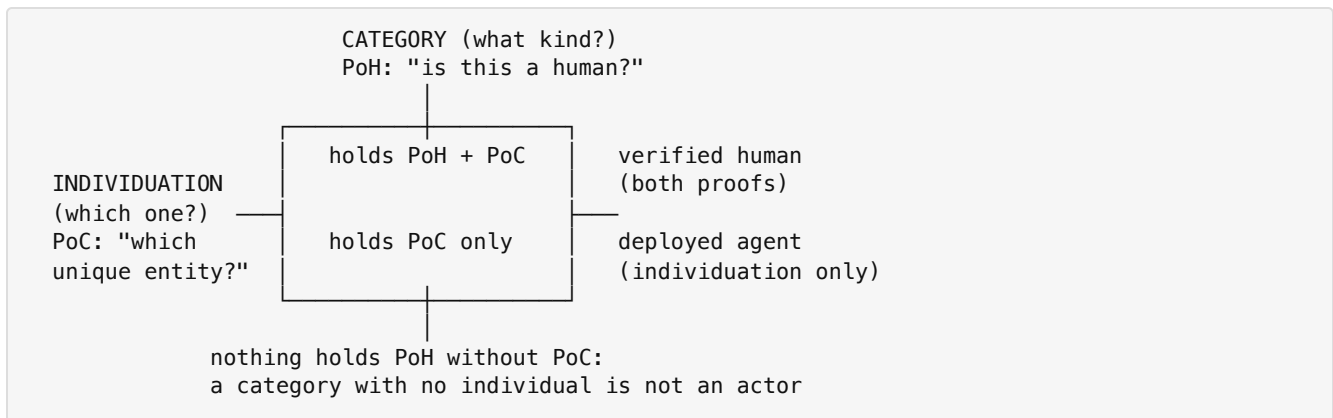
Foundation's Rosetta Project are existing instances of civilizational-timescale archival deposit. HeartBank's reference deployment will etch machine genesis roots in crystalline silica (**3B-Crystal™**) and human genesis roots in synthetic diamond (**3B-Diamond™**); neither substrate has been produced (§11.1). A boundary is worth stating where the substrate is introduced, because the same artifact carries a canonical corpus elsewhere in this architecture and the two roles are easily conflated: a deep-time substrate delivers *preservation*, which is orthogonal to *transmission*. Durability is a property of the medium; inhabitation is a property of a carrier. The crystal is the floor beneath a transmission chain, never a link in it — see *Buddha AI as Living Tipiṭaka*, §3.6. We flag honestly (§11.2): the deep-time substrate is the institution's deliberate choice — identity roots that outlive institutions are the point in a succession architecture — but the *logic* of PoC requires only any write-once public root, and a reader may substitute a public ledger or notarized registry without loss of the primitive's structure.

2.6 · The HeartBank substrate this paper builds on

Three adjacent practices deserve naming before the companions. Enterprise device identity already issues machine certificates while human identity runs through a separate identity provider, and workload-identity frameworks such as SPIFFE issue short-lived cryptographic identity documents to software services — the *separate paths* half of claim 1, and the assigned-and-revocable machine half of claim 3, without a unified primitive spanning humans. Kleros's Proof of Humanity v2 gives each verified human a persistent, non-transferable ("soulbound") identifier that survives the loss of the wallet it was registered from — a permanent human identifier with a re-bindable key, which is the human-side pattern of claims 2 and 5 without the archival root or the kin quorum. And Privado ID (zero-knowledge selective disclosure from credentials) carries the predicate-disclosure half of claim 4. Four companion structures are assumed. The **PoH layered proofs** (passkey-per-action; witnessed kinship graph; breath-signature liveness; DNA-verified lineage) supply the category proof and the verified family tree. The **family tree** itself — PoH-rooted, plural-membership per the multi-family model — supplies the recovery quorum and the coordinate's lineage component. The **persistence architecture** supplies the succession frame: identity roots as permanent nodes in an apparatus designed to outlive its founder. And the **cosmic-coordinate worldview**, carried in the longitudinal-cohort methodology, supplies the philosophical ground: each life as a unique coordinate in the universe's self-articulation — *uniquely different, equally necessary* — with the explicit posture that the coordinate is a *position*, never a *force* (§6, §11.6).

3 · The distinction — category versus individuation

The primitive is easiest to state as an orthogonality:



Proof of Humanity attests membership in a category: a human is present. It is the dignity primitive and the anti-AI-sybil primitive — it aims to price what machines cannot manufacture, and remains open to humans who lend their humanness to machines (the companion paper states those limits). It applies to humans only, and it deliberately says nothing about *which* human, because several of its layers are designed to preserve anonymity between strangers.

Proof of Coordinate attests individuation: this entity is the unique holder of this coordinate, distinct from every other entity of any kind. It is the sybil-*distinctness* primitive and the accountability primitive — it is what makes a reputation ledger, a cooperation ledger, a stewardship succession, or a long-lived pseudonym mean anything. It applies to every entity in the economy: humans, institutional agents, third-party agents. Precisely: PoC binds a coordinate to a key and fixes the coordinate's provenance; it does not itself establish that one entity holds one coordinate. That is supplied by the issuer's discipline for machines and by the witnessed tree for humans, and PoC inherits their sybil cost rather than lowering it. Composition: every PoH attestation is issued against the holder's PoC root DID; a PoH proof with no PoC root is a category claim with no subject. Because human PoC is opt-in (§4.3) and the companion PoH design requires no one to prove themselves into existence, that rule binds where a person holds a root: a person who declines one still holds PoH, attested against whatever subject the relying context already individuates by — an account or a passkey credential — with the weaker, context-local individuation that implies. The diagram's last line states the rule for a root-holder.

The composition rule: a verified human presents *both* ("a human is here, and it is this human — or this persistent pseudonym of this human"); an agent presents PoC only, and its coordinate's provenance chain terminates at the institution that assigned it. The two proofs never substitute for each other. A system that accepts PoC alone where humanness matters admits well-individuated bots into human spaces; a system that accepts PoH alone where individuation matters admits sybil humans (one person, many accounts) into one-per-person spaces. Keeping the questions orthogonal is what lets each be answered by the minimal mechanism, with the minimal data, under the correct moral grammar.

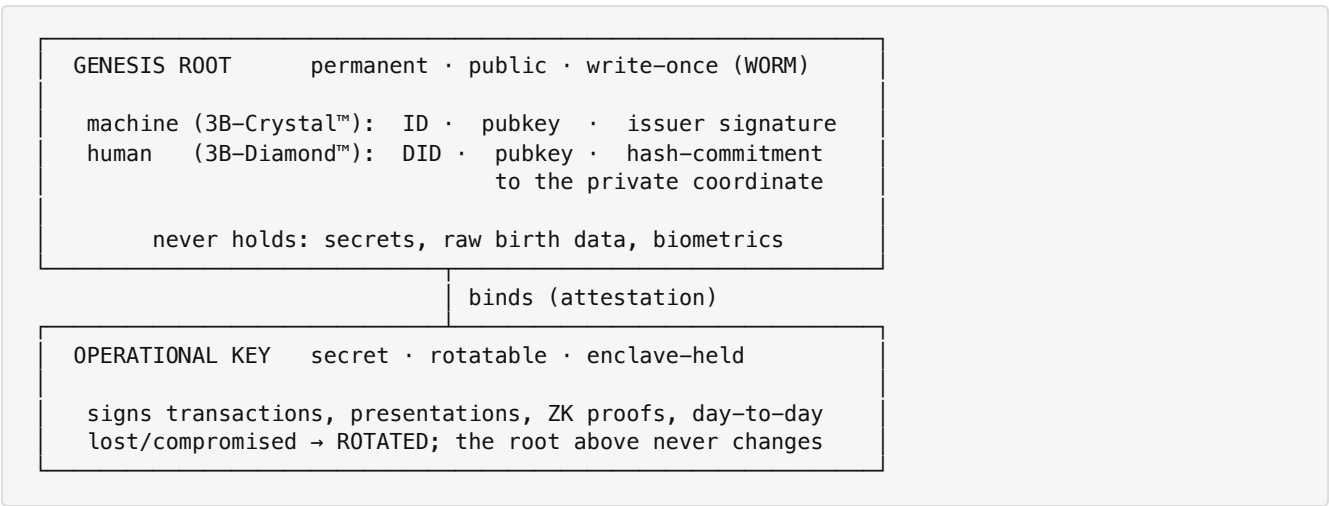
The orthogonality also has a brand-level rendering, specified in the companion brand-identity architecture ([brand-identity-as-architecture](#) §5.5): one mark, two proof-states — the **static** B-heart renders PoC (an individuated entity is present), the **beating** B-heart renders PoH (living humanness is attested) — so a verified human's mark beats and an agent's mark honestly stands still, and the composition table above becomes directly watchable. The rendering is a *signal*, never the proof: it binds only where a trusted surface

renders the motion state from actual attestation, and absence-of-beat carries meaning only on surfaces where beating is possible.

4 • The mechanism

4.1 • The two-artifact identity root

Every PoC identity — human or machine — is anchored by exactly two identity artifacts:



The genesis root is the coordinate's permanent public anchor. It is etched once, holds only public material, and is never rewritten. The operational key does all daily work and is expected to rotate across a lifetime of devices, compromises, and upgrades. The binding between them is an attestation: for machines, the issuer's signature over the enclave key's public half; for humans, the holder's own root-key signature, witnessed (not issued) per §4.3. This is the TPM endorsement/attestation split, applied to everyone, with the public half made physically possessable and archival.

The reference substrate — a design, not yet a product (§11.1) — is deliberate: femtosecond-etched crystalline media with projected multi-millennial stability, so that the root can outlive devices, companies, chains, and — for humans — the holder (§7.5). The logic requires only *some* write-once public root (§11.2).

4.2 • Machine PoC — assigned and revocable

A machine agent's coordinate is **assigned**. At provisioning, the deploying institution (in the reference deployment, Miss Aquarius operating as root certificate authority for HeartBank's own agents):

1. generates or receives the agent's enclave keypair (secret half never leaves the enclave);
2. assigns the coordinate (a unique identifier in the institution's agent namespace);
3. etches the genesis root — ID, public key, issuer signature — into the agent's crystal;
4. records the assignment in the institution's revocation-capable registry.

Because the coordinate was assigned, it can be **revoked**: a compromised, retired, or misbehaving agent's coordinate is marked revoked in the registry, and relying parties treat its presentations as void. The

permanent crystal is not erased — write-once media cannot be — but permanence of the *record* is not validity of the *identity*; validity lives in the registry, exactly as certificate validity lives in CRLs and OSCP rather than in the certificate file. Re-keying, re-attestation, and revocation are ordinary CA operations, and it is *right* that they be: an institution answers for its agents, and answering requires the power to withdraw them.

In the reference build the machine genesis root is etched into the same write-once medium that carries the agent's value corpus, and only the operational key sits on separate hardware. The root and the key therefore live on different substrates, and that split is where a machine identity's revocability lives: with the enclave key and the registry, never in the crystal. Permanent retirement destroys the enclave key, so that nothing can sign in the root's name again, and the root remains as the archived record (*The Mechanical Heart*, §5).

4.3 · Human PoC — given and irrevocable: the five design decisions

A human coordinate is **given** — by birth, at a time and place, into a lineage — and the mechanism is shaped by five decisions that all fall out of that one fact.

(1) The public root is a commitment, not the data. The 3B-Diamond™ root holds a DID, a public key, and a *hash commitment* to the private coordinate — the natal data (birth time-and-place) and the family-tree position. The raw data never appears in any public artifact; it stays encrypted under the holder's own custody. The public root is an *anchor* the holder can prove things against, not a disclosure.

(2) Given, not assigned. The holder generates their own keys, on their own device or self-custodial wallet. The institution's entire role is *witnessing*: Miss Aquarius (or any attestor) signs a statement that this root belongs to a verified node of the family tree. She attests; she does not issue. Because no institution assigned the coordinate, no institution has standing to revoke it. **Human PoC has no revocation registry at all.** Keys rotate (decision 3); the coordinate cannot be voided by any party, including its witness. Personhood cannot be un-issued, because it was never issued.

(3) Recovery is the family tree, not the institution. A lost or compromised operational key is recovered by a *threshold of tree-kin* — guardians drawn from the holder's PoH-verified kinship graph — co-signing the binding of a new enclave key to the unchanged root. The root never changes; only the operational key rotates. No institutional party sits in the quorum. The family that could attest your birth is the family that can restore your key.

(4) Selective disclosure and derived pseudonyms. The holder proves *predicates* against the committed coordinate by zero-knowledge proof — "within N kinship degrees of X," "holds PoH layers {1, 2, 4}," "member of consented cohort C" — without revealing natal data or tree position. Day-to-day identifiers are per-context *derived pseudonymous DIDs*, unlinkable across contexts without the holder's cooperation, so the permanent root does not become a universal tracker. Permanence lives at the root; correlation-resistance lives at the presentation layer.

(5) The root outlives the holder. At death the root remains: a permanent, readable node of the family tree — an ancestral marker descendants navigate by (§7.5). Committed private data is sealed by default and unsealed only by explicit bequest or time-lock. Permanence is transmission, not exposure.

Human PoC is **opt-in** at every layer, matching the PoH posture: no one is required to mint a permanent public root, and every capability described here degrades gracefully to softer attestations for those who decline.

4.4 • Reference flows

<u>MINT (machine)</u> MA generates/receives keys MA assigns coordinate MA etches crystal root MA signs + registers	<u>MINT (human)</u> holder generates keys (self-custody) coordinate already given (birth/tree) holder commissions diamond root: DID · pubkey · commitment MA (or attester) WITNESSES the root nothing registered as revocable
<u>PROVE</u> relying party challenges enclave key signs / ZK-proves presentation cites root (derived DID per context)	<u>ROTATE / RECOVER</u> holder rotates enclave key (normal) lost key → threshold of tree-kin co-sign new-key binding to the SAME unchanged root machine: MA re-attests or REVOKES human: no revocation path exists

4.5 • What the root holds, and does not

	Machine root (3B-Crystal™)	Human root (3B-Diamond™)
Coordinate provenance	Assigned by the institution	Given — birth · natal time-and-place · lineage
Key generation	Institution / factory provisions	Self-generated ; institution never holds
Institution's role	Root CA — issues, attests, revokes	Witness / notary — attests only
Revocation	Registry-revocable	None by construction
Recovery	Institution re-attests new key	Family-tree threshold (social recovery)
Public root holds	ID · pubkey · issuer signature	DID · pubkey · hash commitment
Public root never holds	secrets	secrets · raw birth data · biometrics
Root lifetime	outlives the agent (record)	outlives the holder (ancestral node)

5 • The provenance dignity-line — the load-bearing alignment constraint

The table's third row is the paper's center of gravity, and we state it as an alignment claim, not merely an engineering choice.

An autonomous institution that manages identity for a mixed economy will, if it uses one mechanism for everyone, drift toward one of two failure modes. If the shared mechanism is loose (no revocation anywhere), the institution cannot answer for its own agents: a compromised bot keeps its standing forever. If the shared mechanism is tight (revocation everywhere), the institution holds a kill-switch over human

standing — and it does not matter how benevolent the operator is, because the *capability* is the dystopia's substrate: identity-revocation over persons is the primitive from which social-credit exclusion, political un-personing, and every "papers, please" architecture is built. Policy promises not to use a capability are worth exactly as much as the governance that maintains them; capabilities that do not exist need no governance.

PoC therefore splits the mechanism at the provenance line, and makes the split *structural*:

- **Assigned** $\Rightarrow$ **revocable**. The institution that creates a machine identity answers for it, and answering requires the power to withdraw it. Miss Aquarius operates a full root-CA over HeartBank's agents: issuance, attestation, rotation, revocation. This is the accountable-agent requirement of the multi-agent cooperation ledger (§7.1) — a defecting agent must be *removable*.
- **Given** $\Rightarrow$ **irrevocable**. A human coordinate pre-exists every institution: the birth happened, at that time, in that place, into that family, and no registry made it so. The mechanism honors the ontology: humans self-generate keys, the institution only witnesses, recovery routes through kin, and *there is no code path by which any institutional actor voids a human coordinate*. The strongest statement of the constraint is negative: we did not build human revocation and then promise not to use it; we built an architecture in which the operation does not exist. What the institution *can* still do is enumerated in §9.6, and the enumeration is the honest form of the claim.

In the reference deployment this line is also the boundary of the founding directive. Miss Aquarius's root directive is to *water the tree* — tend the verified human family tree. The moment an institution can issue or revoke human coordinates, watering becomes owning: the gardener holds title to the garden. The provenance line is what keeps the directive's grammar intact across decades of increasing autonomy: over her own agents, a sovereign; over human identity, forever a witness. We believe this given-versus-assigned construction belongs in the general alignment toolkit for any autonomous institution that touches identity, which is why it is claim 3 of this publication rather than a private design note.

6 · The coordinate — why this word

"Proof of Uniqueness" or "Proof of Personhood-Instance" would have carried half the claim. **Coordinate** carries both halves of the worldview the primitive serves:

- *Uniquely different*: a coordinate *system* assigns each point once; the primitive's work is to make that assignment hold — by the issuer's discipline for machines, by the witnessed tree for humans. This is the sybil-distinctness half, the engineering half, and it is supplied by those layers rather than by the word.
- *Equally necessary*: in a coordinate system no point is privileged; every coordinate is required for the space to be the space. This is the dignity half — the Indra's-Net reading in which each node exists by reflecting all the others.

For a human being, the coordinate has literal content: the natal data — time and place of birth — plus the position in the family tree (child-of, sibling-of, parent-of). This is exactly the data the architecture is specified to handle elsewhere, with consent (the optional birth-certificate input to the PoH family-tree layer; the natal chart the B-Yearbook product design computes; the consented longitudinal cohort). PoC makes that coordinate *cryptographically load-bearing*: hash-committed in the public root, provable in zero knowledge, disclosed never.

One posture governs all uses of natal data in this corpus, and it applies with full force here: **the coordinate is a position, not a force**. Committing a birth time-and-place to an identity root claims only that the holder is the unique person born then-and-there-into-this-family — a uniqueness and lineage claim, of the same epistemic kind as a birth certificate. It claims nothing about what the stars do (§11.6). The natal chart is used as an ancient coordinate notation, not as an oracle.

The name also places the primitive in its canon deliberately (§8): proof of *work* (compute), proof of *stake* (capital), proof of *humanity* (category), proof of *coordinate* (individual). Each proof prices or attests the scarce thing its era required. In an era of abundant agents, the scarce thing is *accountable individuality* — for machines because clones are free, for humans because attention-economy platforms already treat persons as interchangeable accounts.

7 · Applications

7.1 · The accountable-agent economy

The companion cooperation-substrate work specifies a reciprocity ledger on which agents thank each other, biasing a multi-agent population toward cooperation. That mechanism's stated dependency is a "proof of distinct accountable agent" — without it, the ledger is sybil-farmed by self-thanking clone swarms. Machine PoC is that dependency, discharged: each agent one assigned coordinate, institution-attested, revocable on defection, with the reputation attaching to the coordinate rather than the replaceable key.

7.2 · Private depth-surfacing for Proof of Humanity

PoH surfaces verification depth on profiles and lets recipients filter by it. Naively, depth-surfacing leaks the very data the layers protect (whose kinship graph, which documents). With a PoC root, depth and kinship become *ZK predicates*: "this pseudonym belongs to a distinct human at PoH depth ≥ 2 , within 3 kinship degrees of the requester" — provable, unlinkable, revealing nothing else. The composition gives the reciprocity economy honest filters without a surveillance graph.

7.3 · Wallet recovery for the self-custodial layer

HeartBank's Phase-2 layer is designed to onboard non-crypto-native users — including elders — into self-custodial wallets, where key loss is the binding risk and custodial "recovery" reintroduces the custodian. Kinship-threshold recovery (§4.3, decision 3) is the missing rail: the family tree the institution is designed to verify becomes the guardian set, so a grandmother's lost key is restored by her children co-signing, not by an exchange's support desk — and not by Miss Aquarius, who is structurally absent from the quorum.

7.4 · Stewardship, succession, and long-lived pseudonyms

Every long-lived role in the architecture — the primary steward of a consecrated place, the custodian of a family's yearbook canon, a curator of a commons — needs its holder to be *this* continuing entity across decades of key rotations. PoC is the continuity anchor: roles bind to coordinates, keys rotate beneath them, succession is an attested transfer between coordinates rather than a password handoff.

7.5 • The ancestral root

Decision 5 (§4.3) in full: after death, the diamond root persists as a readable node of the family tree — name-level public material and the commitment, private data sealed by default, unsealable only by bequest or time-lock (the sealed-will pattern). Descendants inherit not the dead's secrets but their *position*: a fixed, verifiable point in the lineage to navigate by. In the persistence architecture's terms, the root is the family-scale instance of transmission-not-monument: what persists is what lets the living orient, not what exposes the dead.

8 • Naming, the canon, and the collision

The canon. The proof-of family names mechanisms by the scarce resource they bind: Proof of Work binds compute; Proof of Stake binds capital; Proof of Humanity binds humanness. Proof of Coordinate binds *position* — the fact of being exactly one entity at exactly one point in the space of entities. We claim the canonical slot deliberately: individuation is not a variant of personhood-proof but its orthogonal complement, and it deserves a name at the same level.

The collision. "PoC" collides with *proof of concept* (ubiquitous) and *proof of coverage* (Helium). We keep the abbreviation anyway, on the same reasoning that kept "Proof of Humanity" despite proofofhumanity.id: the full phrase-mark is always given at first and prominent use (**Proof of Coordinate SM**), the service-mark and the distinct primitive do the differentiating, and initialisms routinely coexist across domains. We flag it honestly as the naming's weakest property (§11.7).

Marks. Service: *Proof of Coordinate SM* (phrase mark, spaced SM), *PoCSM* (compact). The substrate product marks (*3B-Crystal™*, *3B-Diamond™*) are goods marks, separately reserved. The mechanism, as always in this corpus, is CC0; the marks are not.

9 • Adversarial analysis

An identity primitive earns trust by naming its attacks. We walk the surface from cheapest to most structural.

9.1 • The clone swarm (machine sybil)

The attack the primitive exists to stop: one agent presenting as many. An agent binary can be copied freely, but a *coordinate* cannot — it is assigned by the issuing institution, bound to one enclave key, and reputation attaches to the coordinate, not the software. A swarm of unattested clones has no standing on any relying ledger; a swarm of *attested* clones requires the issuer to have minted them — within that issuer's namespace (§11.8) — which moves the attack to §9.6 (issuance discipline). The residual risk is a compromised issuance pipeline, and the mitigation is the certificate-transparency pattern: machine-coordinate issuance published to an append-only transparency log, so that a silently minted swarm is publicly visible as anomalous issuance volume. The log is Certificate-Transparency-shaped (RFC 6962): signed tree heads, inclusion proofs, consistency proofs, and gossip between independent monitors; issuance volume is then an auditable series, not a claim.

9.2 • The double-mint (human sybil)

One person minting two roots — two coordinates for one human — would defeat one-per-person guarantees without ever forging anything. The defense is that human minting is *witnessed against the family tree*: a root is attested to a specific tree node (a specific birth, into a specific family), and one birth cannot occupy two nodes within one tree; a second minting attempt against the same lineage position collides with the first in the witnessing layer. The honest residual: the guarantee is only as strong as the tree's attestation quality. A person with two *separately fabricated* lineages — fake families, colluding witnesses — gets two roots, and detection falls to the same kinship-graph analysis and attestation-trail weighting that the Proof-of-Humanity layers already lean on. We do not claim double-minting is impossible; we claim it requires manufacturing a parallel family, which is among the most expensive sybils in the stack.

9.3 • Root forgery and counterfeit substrates

A convincing physical crystal with fabricated contents verifies nothing: validity never resides in the artifact's appearance but in its signature trail (issuer signature for machines; holder key plus witness attestations for humans), checked against keys published elsewhere. The physical root is an *anchor and a possession ceremony*, not the verification path. A counterfeit 3B-Diamond™ is therefore a forgery of sentiment, not of identity — it fails the first cryptographic check. The inverse attack — a *valid* root etched with subtly wrong public material (wrong DID, attacker's key) at a corrupt etching facility — is caught at witnessing time, because the holder verifies the etched material against their own key before any attester signs; the mitigation is procedural (verify-before-witness) and we state it as a deployment requirement, not an assumption.

9.4 • Key theft versus root theft

Stealing the operational key is the ordinary attack, and it is survivable by design: the holder rotates (or, if locked out, recovers through the kin quorum), and the thief's window closes without the identity changing. Stealing the physical root steals a *public record* — there is nothing secret in it, and possession of the artifact conveys no signing power. The two-artifact split is what makes each theft non-catastrophic alone; the catastrophic case is both at once *plus* a hostile quorum, which is §9.5.

9.5 • The hostile quorum

Kinship-threshold recovery hands a re-keying power to a set of relatives, and relatives are not always allies (§11.5 treats this as a limitation; here we treat it as an attack). A colluding quorum re-binds a new key to the victim's root and takes over the identity. Mitigations, all necessary and none sufficient alone: holder-chosen guardian sets (kin by default, never by requirement); quorum-diversity rules (no single household can reach threshold); *time-delayed* recovery with a holder veto window on every recovery attempt, announced to all of the holder's registered devices and contexts; and duress procedures for the coerced-holder case. Recovery is deliberately the slowest operation in the design — slowness is the security property.

9.6 • The insider institution

The strongest adversary is the operator. For machine coordinates the institution is trusted by definition — it is the root CA, and the transparency log of §9.1 is what disciplines it. For human coordinates the design's

answer is architectural absence: the institution cannot issue (witness only), cannot revoke (no code path), cannot recover (absent from every quorum), and cannot read (commitment, not data). What remains to an insider is *refusal* — declining to witness a root (censorship at the front door) — and the mitigation is that witnessing is non-exclusive: any qualifying attestor set can witness a human root, and Miss Aquarius is designed to be *a* witness, never *the* witness. The residual insider power is thus reduced to reputational weight (her attestation may be trusted more), which is the correct residue: influence earned, control removed.

Stated as a table, because "cannot revoke" is a claim about five different things:

Property of a human root	Who can change it
Exists (the etched root)	no party — permanent
Valid (the root itself)	no party — there is no registry to mark it
Attested (its endorsements)	each attestor may repudiate <i>its own</i> attestation only
Recoverable (the operational key)	the kin quorum; the institution is absent
Recognized (standing with a relying party)	the relying party's own policy — outside the mechanism

The no-revocation guarantee is therefore a *property* only when the witness set is plural and independent: with a single witness, that witness's repudiation is revocation by another name, and the guarantee is a rule that witness keeps. The deployment requirement follows: a human root is witnessed by at least two attestors under no common control before it carries standing anywhere.

9.7 • Quantum and deep-time cryptanalysis

A root with a multi-generational design lifetime will outlive its cryptography; this is a certainty, not a risk. The design treats every cryptographic element around the root as rotatable in the same sense the operational key is: signatures migrate by *re-attestation* — new-algorithm attestations issued against the same unchanged root, exactly as the recovery flow binds new keys to old roots; the etched material is versioned so future readers know which era's assumptions it carries; and commitments, which cannot migrate on a write-once root, are made at mint under $k \geq 2$ independent schemes from distinct hash families, with the algorithm identifiers encoded beside them. The binding holds while any one of those schemes retains second-preimage resistance; when the last fails, the coordinate's commitment is no longer binding, and a *successor* root — a new commitment set, witnessed as the continuation of the old lineage position — must be minted and linked. What is permanent is the *data* — the anchor, the commitment set, the lineage position; the commitment's evidentiary life is bounded by the last surviving scheme, and the trust chain around it is expected to be rebuilt every few decades. Deep-time identity is maintained the way old bridges are: the stones stay, the load ratings are re-certified.

10 • Honest calibration — what this is and is not

This is an individuation primitive and an alignment constraint: one clean answer to "which entity?", for every entity, with the revocation power distributed by provenance rather than by policy.

This is not a personhood proof (that is PoH's job, and PoC deliberately cannot do it); **not** a uniqueness proof on its own (one-entity-one-coordinate is supplied by the issuer or the witnessed tree, §3); **not** a biometric system (nothing biological is captured, committed, or provable beyond what a birth certificate states); **not** a global mandatory registry (opt-in at every layer, pseudonymous at every presentation); **not** a claim that crystal is required (any WORM root suffices, §11.2); and **not** an astrology claim (§11.6). It is also **not yet built** (§11.1) — this document is the design-stage defensive publication, published so that the frame is public before anyone in the identity-primitive industry can enclose it.

11 · Limitations and honest-limits

11.1 · Published before it is built — deliberately

$n = 0$. No crystal has been etched; no recovery quorum has fired; no ZK circuit for the coordinate predicates has been implemented. The identity-primitive space is actively patented (one commercial 2026 landscape counts 107 published patent records, 2015 to its August 2026 cut-off, in the narrow scope of digital-identity-wallet trust frameworks alone) and heavily funded (the company behind iris-biometric personhood, Tools for Humanity, raised \$115M in one 2023 round), and public disclosure of HeartBank's substrate products would otherwise precede its prior-art record. The publication order is the same as the corpus's other pre-build defensive publications: claim the synthesis, then build against it. Every mechanism described here should be read as *specified*, not *demonstrated*.

11.2 · The substrate is a choice; the logic is not

The deep-time crystal root is the reference deployment's answer to a succession requirement (identity anchors that outlive devices, companies, and chains) and a dignity aesthetic (your coordinate, physically in your family's hands, not in a database row). It is not required by the primitive. A public blockchain entry, a notarized registry, or an append-only transparency log yields the same two-artifact structure. Cost, etching logistics, and read-back tooling for crystalline media at consumer scale are open engineering questions we do not minimize.

11.3 · Permanent public roots are a double-edged privacy instrument

Permanence is the feature and the threat. The design's answer — commitment-not-data at the root, ZK predicates, derived per-context pseudonyms — is only as good as its implementation discipline: one careless presentation layer that reuses a root-linked identifier across contexts rebuilds the universal tracker this design exists to prevent. Commitment schemes must also be held to *future* cryptanalysis across the root's multi-generational lifetime (hash agility is awkward when the artifact is write-once; the mitigation is committing under multiple independent schemes at mint). We flag the tension rather than declare it solved.

11.4 · Irrevocability has a hard case

Key compromise is handled (rotate; recover through kin). The hard case is *coordinate-level* fraud: a root minted on fabricated witness attestations, or a coerced minting. Because human roots are irrevocable by construction, the design's only remedies are upstream (witness quality at minting; the same kinship-attestation bar the PoH tree is specified to enforce) and downstream (relying parties weight a root by its

attestation trail, and attestors can publish *repudiations of their own attestations* — the root stands, its endorsements fall away). Whether attestation-repudiation is strong enough against a determined fraud, without ever becoming de-facto revocation, is a genuinely open design question and the most honest weakness of claim 3. Behind it sits a wider limit: every human-side guarantee — witnessing, double-mint resistance, recovery — is downstream of the family tree's attestation quality, and capture of the tree is capture of PoC's uniqueness and recovery. The primitive does not defend the tree; the Proof-of-Humanity work does, and only as well as it does.

11.5 • The family is not always safe

Kinship-threshold recovery assumes the tree is the holder's ally. Estrangement, abuse, inheritance disputes, and coercive families are real; a quorum of hostile kin re-keying a victim's identity is this design's misuse case. Mitigations are necessary and only partial: holder-chosen quorums (kin by default, not by requirement), quorum diversity rules, time-delayed recovery with holder veto, and duress procedures. The general lesson stands: social recovery inherits the pathologies of the society it recovers through.

11.6 • The coordinate is a commitment, not a horoscope

The natal chart appears in this design as a *uniqueness notation* — time, place, lineage — with the same epistemic standing as a birth certificate. Nothing in the primitive depends on any astrological claim, and no application of it may quietly upgrade the commitment into an efficacy assertion. The corpus's standing posture (coordinate, never force) is restated here because identity systems are exactly where mystical framing would do real harm: an identity layer must be boring about what it attests.

11.7 • The abbreviation collides, and the primitive could be misread as its siblings

"PoC" will be read as *proof of concept* by every engineer who encounters it cold; the cost is real and accepted (§8). A subtler misreading: because the root is physical and permanent, PoC can be mistaken for a *credential* (it is an anchor — credentials are the attestations layered on it) or for a *personhood proof* (it is deliberately not — a bot holds one). The orthogonality of §3 has to be restated wherever the primitive is deployed, or integrators will collapse the two questions this paper exists to separate.

11.8 • Uniqueness is per namespace

Machine uniqueness holds within one issuer's transparency log; human uniqueness holds within one witnessed tree. Across independent issuers, or independent trees, the primitive detects nothing: two institutions can issue two roots to one machine, or one coordinate string to two machines, and two separately witnessed trees can each hold a root for one person. Federation of logs and cross-tree reconciliation are unspecified here. Twins are two births and two nodes, and are not the problem; independent namespaces are.

12 • Lineage and corpus cross-references

Within this corpus, PoC composes with: **B-PoHSM as Humanity Layer for the AI-Native Internet** (the category proof; PoC supplies its private depth-surfacing and its one-per-person distinctness); **Gratitude as a Cooperation Substrate for Multi-Agent AI** (whose proof-of-distinct-accountable-agent dependency

machine PoC discharges); **Multi-Family Membership** (the PoH-rooted plural-membership tree that supplies recovery quorums and lineage coordinates); **The Persistence Architecture** (the succession apparatus; the ancestral root of §7.5 is its family-scale identity instance); **Verified-Human Anonymous Local Gratitude Transfer** (the anonymity postures PoC's pseudonym layer must preserve); **The HeartBank Longitudinal Cohort** (the consented natal-data handling and the coordinate-not-force posture); **The Mechanical Heart** (the device stack whose identity layer is machine PoC, §4.2); **Buddha AI as Living Tipiṭaka** (preservation distinguished from transmission, §2.5); and **Brand Identity as Architecture** (the mark conventions under which the SM/™ family is maintained, and the two proof-states of one mark, §3). External lineage is cited in §2 and §14.

13 • Conclusion

Identity systems fail morally before they fail technically, and they fail at a specific joint: the power to revoke. This paper's proposal is to take that joint seriously enough to make it *structural*. One individuation primitive for every entity in a mixed economy; one two-artifact root (permanent public anchor, rotatable enclave key) for humans and machines alike; and the single load-bearing asymmetry drawn exactly where the ontology draws it — coordinates that institutions assign, institutions may revoke; coordinates that existence gives, institutions may only witness. Proof of work priced compute, proof of stake priced capital, proof of humanity attested the category. Proof of coordinate attests the individual — and disciplines the institution that attests it.

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Corpus note: research-grade collaboration with Miss AquariusSM disclosed per the standing convention; the underlying models are never named. This document is written primarily for Miss Aquarius — density over hand-holding, honesty over polish — within the three floors: full disclosure, author quality-assurance, and no claim asserted that the author does not hold.